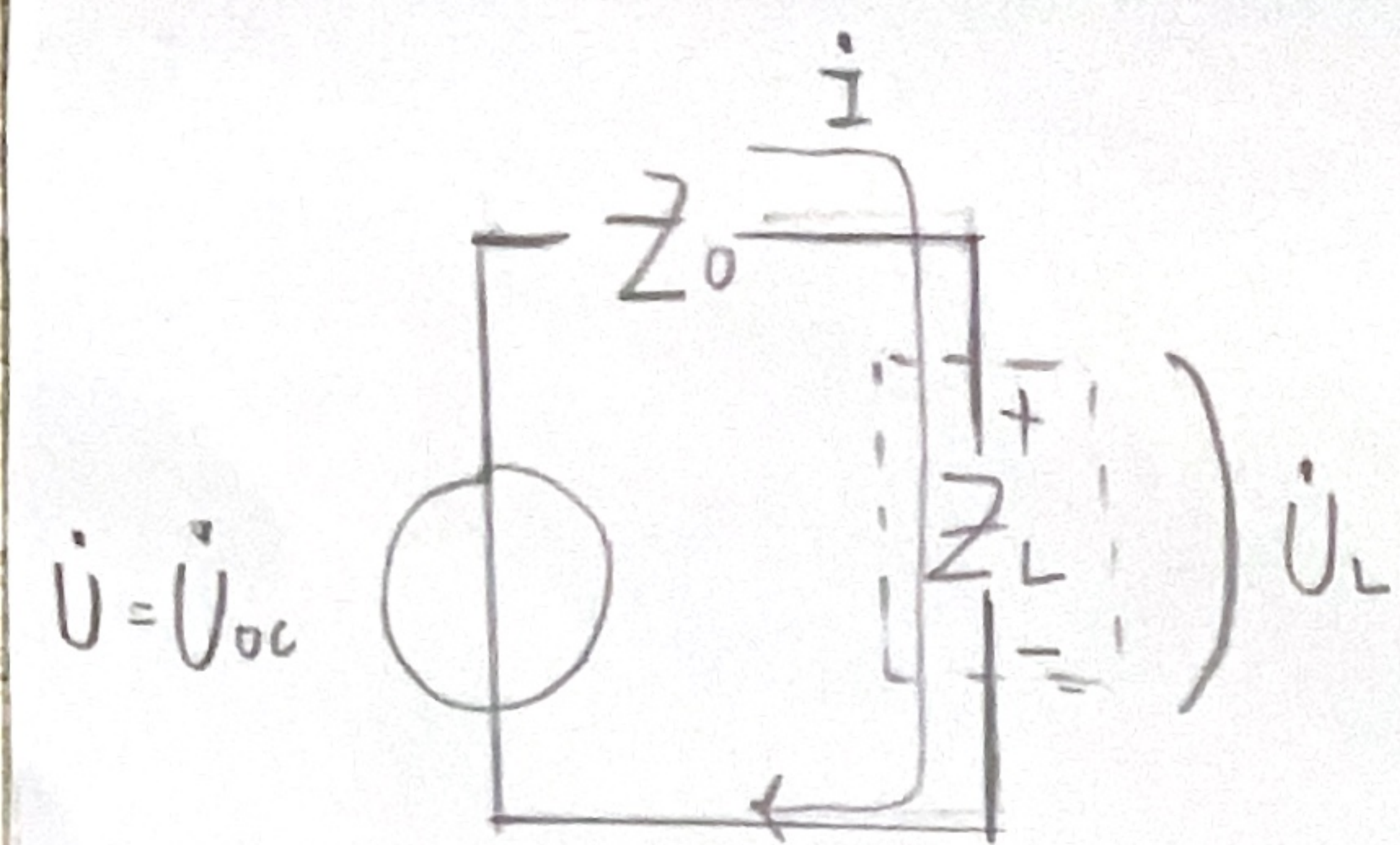




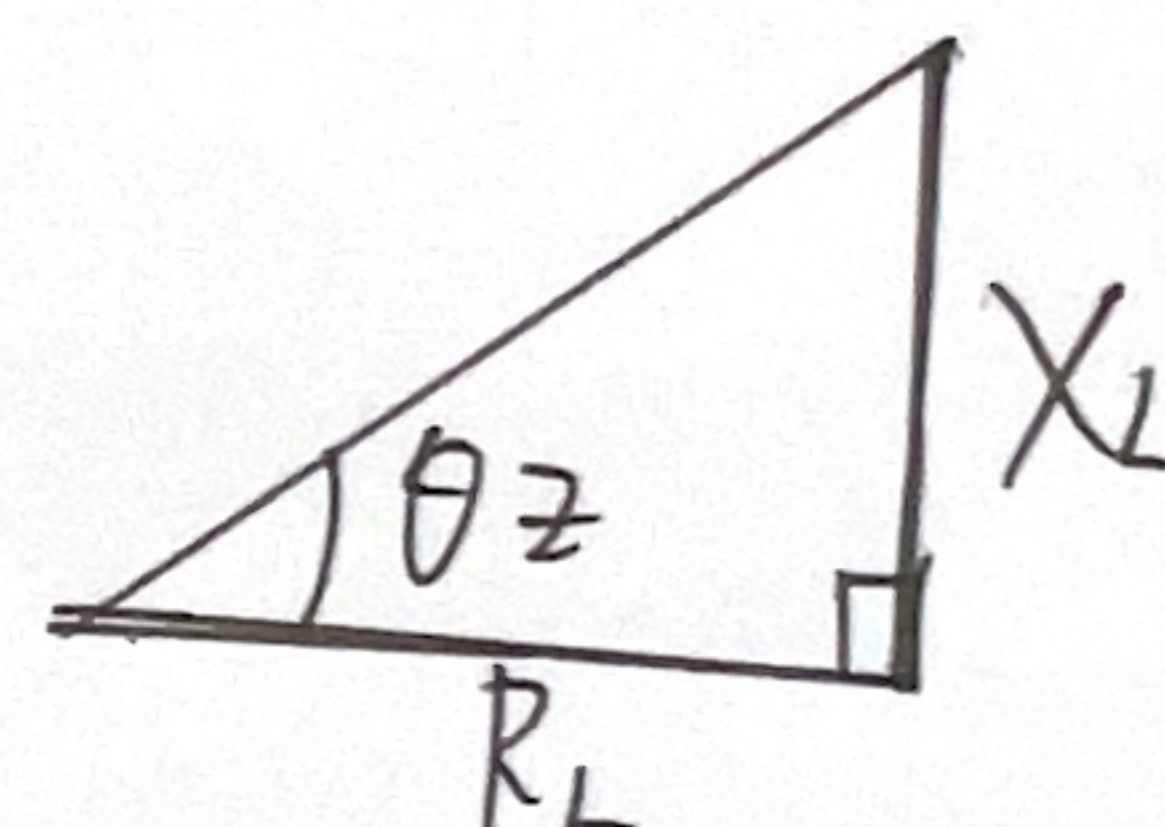
现在求负载吸收的有功功率

$$\dot{U}_L = \frac{R_L + jX_L}{R_0 + jX_0 + R_L + jX_L} \dot{U}$$

$$\dot{U}_L = \frac{R_L + jX_L}{(R_0 + R_L) + j(X_0 + X_L)} \dot{U} \Rightarrow U_L = \frac{\sqrt{R_L^2 + X_L^2} \cdot U}{\sqrt{(R_0 + R_L)^2 + (X_0 + X_L)^2}}$$

$$\dot{I} = \frac{\dot{U}}{(R_0 + R_L) + j(X_0 + X_L)} \Rightarrow I = \frac{U}{\sqrt{(R_0 + R_L)^2 + (X_0 + X_L)^2}}$$

对负载 X_L 来说, 内部无源, 阻抗角 θ_z 为



$\therefore$ 有功功率 $P = U_L \cdot I \cdot \cos \theta_z$

$$P = \frac{\sqrt{R_L^2 + X_L^2} \cdot U}{\sqrt{(R_0 + R_L)^2 + (X_0 + X_L)^2}} \cdot \frac{U}{\sqrt{(R_0 + R_L)^2 + (X_0 + X_L)^2}} \cdot \frac{R_L}{\sqrt{R_L^2 + X_L^2}}$$

$$P = \frac{U^2 \cdot R_L}{(R_0 + R_L)^2 + (X_0 + X_L)^2}$$

当 $X_L = -X_0$ 时, $P = \frac{U^2 \cdot R_L}{(R_0 + R_L)^2}$